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Chapter 1

class notes

1.1 Coulomb's Law

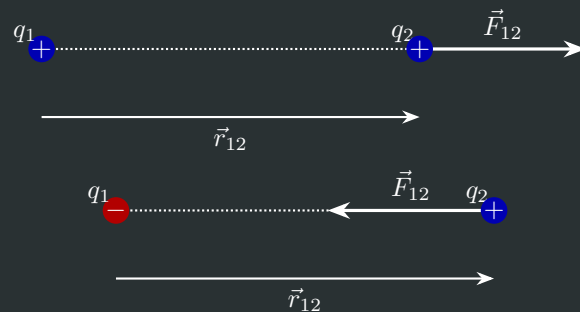
Coulomb's Law

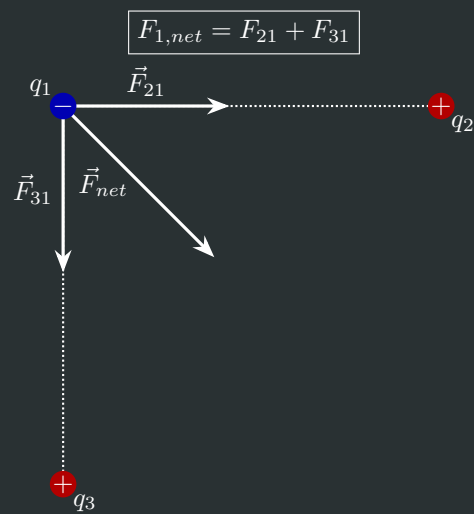
$$\vec{F}_{12} = k \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12} \quad (1.1)$$

Gives a force along the line of two charges.

If q_1 and q_2 have the same sign (i.e. charge) then the force must be repulsive. But if they have the *same* sign, then it must be attractive.

Because of Newton's second law, $F_{12} = -F_{21}$





Superposition Principle

$$F_{net} = \sum_i \vec{F}_i \quad (1.2)$$

Electron Charge

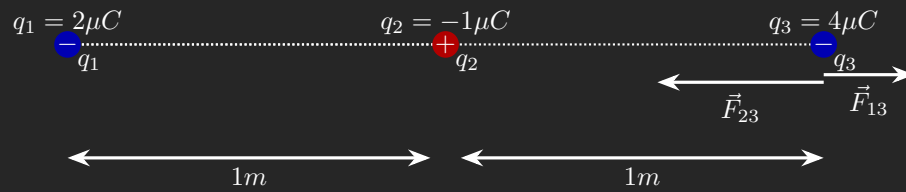
$$e = -1.6 \times 10^{-19} C$$

k

$$k = 9 \times 10^9 \frac{NM^2}{C^2}$$

1.1.1 Examples

Example 1.1.1 $F_{3,net} = ???$



Using Coulomb's Law 1.1...

$$\vec{F}_{3,net} = \vec{F}_{31} + \vec{F}_{32}$$

$$\vec{F}_{31} = k \frac{q_3 q_1}{r_{31}^2} \hat{r}_{32}$$

$$\vec{F}_{32} = k \frac{q_3 q_2}{r_{32}^2} \hat{r}_{32}$$

$$k = 9 \times 10^9 \frac{NM^2}{C^2}, \quad q_1 = 2\mu C, \quad q_2 = -1\mu C, \quad q_3 = 4\mu C$$

$$\vec{F}_{31} = 1.800 \times 10^{-2} \text{ N}, \quad \vec{F}_{32} = -3.600 \times 10^{-2} \text{ N}$$

$$\vec{F}_{3,net} = -1.800 \times 10^{-2} \text{ N}$$

1.2 Electric flux and Field Lines

1.2.1 Field Lines

- Measure of electric field lines over an area

Electric Field

$$E = k \frac{q}{r^2} \quad (1.3)$$

Measure of the energy in an electric field at a point

Density of Field Lines

$$D = \frac{N}{4\pi r^2} \quad (1.4)$$

used when plotting

Number of Graphed Field Lines

$$N \equiv \frac{q}{\epsilon_0}, \quad E = \frac{N}{A} \quad (1.5)$$

1.2.2 Flux

Number of field lines passing through a surface

$$\Phi \propto N$$

For a point charge aligned at the center of a sphere it can just be defined as

$$\Phi \equiv EA$$

To calculate total over surface ...

$$\Phi \equiv \int_s \vec{E} d\vec{A}$$

1.3 5: potential energy

Guass's Law

$$\int E dS = \frac{Q_{enclosed}}{\epsilon_0} \quad (1.6)$$

Change in potential in const E field

$$\Delta U = q\vec{E}\ell \quad (1.7)$$

Change in potential in non static field

$$\Delta U = Q \int_a^b \vec{E} dr \quad (1.8)$$

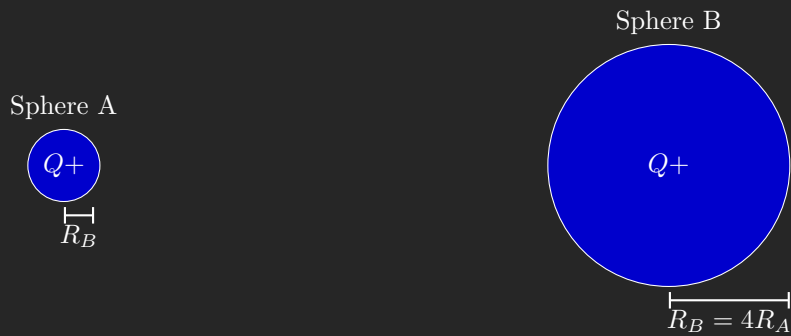
1.7 Conductors and Capacitance

1.7.1 Conductors

- Every point in a conductor is at equipotential, this is because the electric field at any point is zero.
- Continuing with Gauss's law, no matter the location of a charge contained *inside* a shell (but not overlapping the shell itself), the outside of the shell will have the same charge distribution, however the inside may not.
- When looking at the inverse, (a particle placed *outside* of a conductor), we find that the charges move to the surface to cancel out the exerted field, meaning that the inside of the object will have a net charge of 0.
- Importantly, this still applies when a cavity is made *within the conductor*
- This allows for the shielding of an object if it is placed within a conductor ¹

¹Think Faraday cage (maybe)

Example 1.7.1 Equipotential difference in two Spheres



$$V_A = - \int_{\infty}^{R_A} \vec{E}_A \cdot d\vec{l}, \quad V_B = - \int_{\infty}^{R_B} \vec{E}_B \cdot d\vec{l}$$

Given a point charge ...

$$\vec{E} = k \frac{Q}{r^2} \rightarrow - \int_{\infty}^r k \frac{Q}{r^2} = k \frac{Q}{r} \rightarrow$$

For a point charge...

$$V = k \frac{Q}{r}$$

$$V_A = k \frac{Q}{R_A}, \quad V_B = k \frac{Q}{4R_B}$$

$$4 \times V_A = V_B$$

Once connected, become a single conductor which wishes to reach equipotential. To find final charges ...

$$V_A = V_B \Rightarrow k \frac{Q_A}{R_A} = k \frac{Q_B}{R_B}$$

$$k \frac{Q_A}{R_A} = k \frac{Q_B}{4R_A}$$

$$4Q_A = Q_B$$

1.7.2 Capacitance

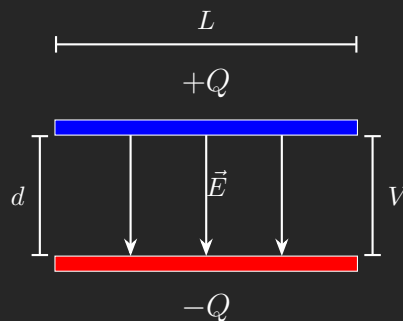
Capacitance

$$C \equiv \frac{Q}{\Delta V} \quad (1.9)$$

Units of Farads, or Coulombs per volt

- Measure of field between to oppositely charged objects
- Ex. A plate of Q and $-Q$ placed a distance d away from one another
- Stores the energy *in* the magnetic field induced between said objects

Example 1.7.2 Capacitance between two plates



$$d \ll L$$

To calculate E field between plates ...

$$E_{bot} = \frac{1}{2} \frac{\sigma}{\epsilon_0}, \quad E_{top} = \frac{1}{2} \frac{\sigma}{\epsilon_0}$$

$$\sigma = \frac{Q}{A}$$

$$E_{bot} = \frac{1}{2} \frac{Q}{\epsilon_0 A}, \quad E_{top} = \frac{1}{2} \frac{Q}{\epsilon_0 A}$$

Since field lines are the same direction...

$$E_{tot} = \frac{Q}{\epsilon_0 A}$$

To find capacitance...

$$C \equiv \frac{Q}{\Delta V}$$

$$|\Delta V| = \int_{bot}^{top} \vec{E} \cdot d\vec{l} \Rightarrow \Delta V = \frac{Q}{\epsilon_0 A} \int_{bot}^{top} dl$$

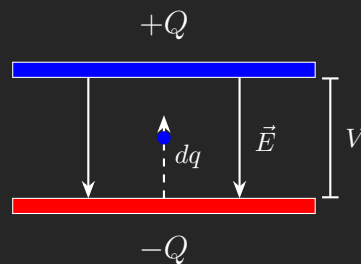
$$\Delta V = \frac{Q}{\epsilon_0 A} d$$

$$C = \frac{\epsilon_0 A}{d}$$

Capacitance for Parallel Plates

$$C = \frac{\epsilon_0 A}{d} \quad (1.10)$$

Example 1.7.3 Parallel Plate Capacitor



To calculate the work needed to move a particle from the bottom plate to the top plate

$$dW = dqEd$$

Since we know that the energy from the electric field is equal to $\frac{V}{d} \dots$

$$dW = dq \frac{V}{d} d$$

$$dU = Vdq$$

This holds true for *any system*. Note that dU is *not* constant because for each particle moved, it becomes requires more work for the next.

To calculate potential of a capacitor. . .

$$U = \int_0^Q V dq \Rightarrow U = \int_0^Q \frac{q}{C} dq$$

Thus . . .

$$U = \frac{1}{2} \frac{Q^2}{C}$$

Potential of a Capacitor

$$U = \frac{1}{2} \frac{Q^2}{C} \quad U = \frac{1}{2} CV^2 \quad U = \frac{1}{2} QV \quad (1.11)$$

Energy density of a capacitor

$$u = \frac{1}{2} \epsilon_0 E^2 \quad (1.12)$$

1.8 Capacitors

What happens if we pull apart a parallel plate capacitor?

- Q remains constant
- E increases
- ΔV increases
- C decreases
- U increases

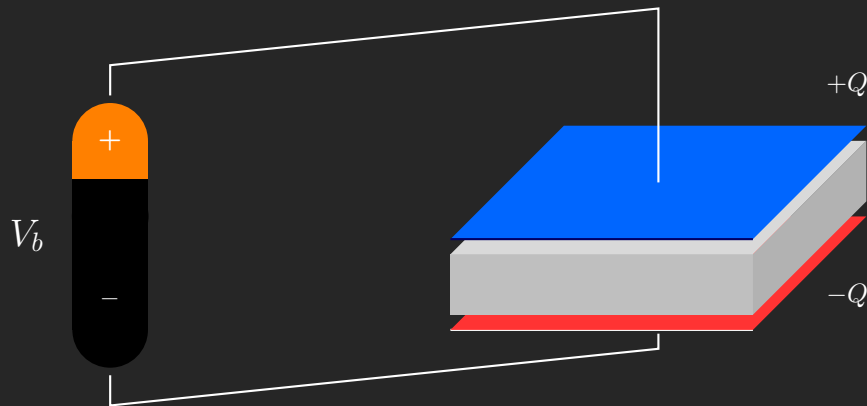
1.8.1 Dielectrics

- Material where positive and negative charges in individual molecules can change, but not through the entire material
- When placed between a capacitor all of the charges align themselves to the field
- This creates a material full of little dipoles
- This induced field is the *opposite* of the one from the plates
- This has the effect of *reducing* the electric field within the capacitor, which then reduces the potential diff (V), which then *increases* the capacitance

$$C = \frac{Q}{V}$$

- The amount a capacitance increases is called the **Dielectric Constant** and is calculated via $C_{old} = \kappa C_{new}$

Example 1.8.1 Adding a Dielectric to a Capacitor



- Capacitance increases
- $V_c = V_b$
- Because $C = \frac{Q}{V} \Rightarrow Q \uparrow$
- Because $U = \frac{1}{2} Q V_c \Rightarrow U \uparrow$

1.8.2 Capacitors in Parallel

- When wired in parallel they can be treated as one capacitor with a larger area
- this equivalent capacitor has...

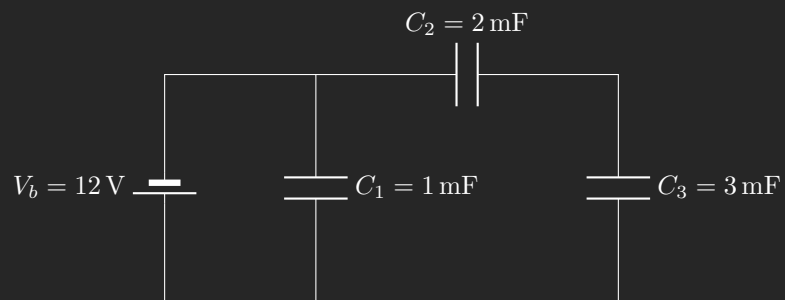
$$V_{equiv} = V_1 = V_2 \quad C_{equiv} = C_1 + C_2, \quad Q_{equiv} = Q_1 + Q_2$$

1.8.3 Capacitor in series

- When wired in series they act like a capacitor with a large distance
- This equivalent capacitor has ...

$$Q_{equiv} = Q_1 = Q_2 \quad \frac{1}{C_{equiv}} = \frac{1}{C_1} + \frac{1}{C_2}, \quad V_{equiv} = V_1 + V_2$$

Example 1.8.2 Combination of Capacitors



Simply into single capacitors

$$C_2, C_3 \rightarrow \frac{1}{C_{23}} = \frac{1}{C_2} + \frac{1}{C_3} = 1.2 \text{ mF}$$

$$C_1, C_{23} \rightarrow C_1 + C_{23} = C_{123} = 2.2 \text{ mF}$$

Chapter 2

disc

2.1 pset1

2.1.1 a

It is possible that the third option c could be true. This is because we see an exponential acceleration in the Y direction, which correlates with W^2

2.1.2 b

$$a = F \quad t = \frac{L}{V_0} \rightarrow y_f = \frac{1}{2} \frac{FL^2}{V_0}$$

2.1.3 c

$$\frac{w}{2} = \frac{1}{2} \frac{F_{max} L^2}{mv_0^2} \quad F_{max} = \frac{mv_0^2 w}{L^2}$$

2.1.4 d

$$\tan(\theta) = \frac{v_y}{v_o} = \frac{w}{l}$$
$$v_y = \frac{F}{m} \times \frac{L}{v_0}$$

2.2

2.2.1

It has a net force of 0. This is because the x and y components cancel each other out.

2.2.2

The y component would cancel, as it is symmetric around the particle. X would not form a triangle to find r to use $u_g = \frac{-Gm_1m_2}{r^2}$

$$F_g = -\frac{GMm}{\left(\frac{\sqrt{5}a}{2}\right)^2}$$

$$F_{g,x} = -\frac{16Mm}{g\sqrt{5}a^2}$$

2.2.3

2.2.4

The physical argument is that the object is quite far away, and thus we can use a small angle aprox.

$$\sqrt{(x - a/2)^2 + (a/2)^2} \rightarrow \approx x$$

$$\frac{GMm}{x^2}$$

2.3 RL stuff lol

it is the fourth one

2.3.1

$$u_c = -\frac{4\sqrt{2}GMm}{a}$$

2.3.2

$$\Delta u = \frac{4\sqrt{2}GMm}{a} = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{8\sqrt{2}GM}{a}}$$

2.3.3

$$u_1, u_2 = -\frac{2GMm}{a}$$

$$u_3, u_4 = \frac{-2GMm}{\sqrt{5}a}$$

$$u = u_1 + u_2 \cdots = -\frac{4GMm}{a}\left(1 + \frac{1}{\sqrt{5}} = Ur\right)$$

2.3.4

no, the middle has *higher* potential energy

2.3.5

2.4

2.4.1

$$p^2 = \frac{4a^3\pi^2}{GM}$$

2.4.2

$$\omega = 2\pi/p \rightarrow \omega^2 = \frac{4\pi^2 GM}{\cancel{4a^3\pi^2}} = \frac{GM}{r^3}$$

2.4.3

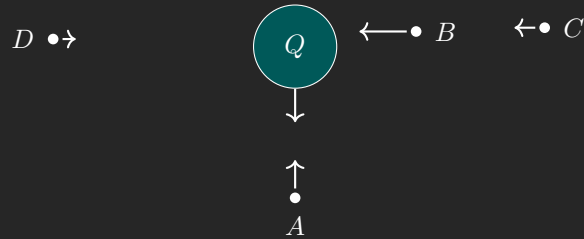
$$\begin{aligned} Dw^a R6a + b \\ a = \frac{D}{m} w^a R^{a+b} = \omega^2 R \\ \omega^{2-a} = \frac{D}{m} R^{a+b-1} \end{aligned}$$

2.4.4

2.5 w2

2.5.1 q1

Example 2.5.1 The configuration below shows some points around a positive charge Q . If a positive test charge q is placed at point A, at a distance r_A from Q , calculate the magnitude of the force between the charges in terms the given variables.



P1) Find magnitude

Because we're finding a charge diff, we use 1.1

$$\vec{F}_{12} = k \frac{Qq}{r_A^2}$$

P2) Force strength

Based on distance to the *field* of Q

$$F_B > F_A > F_C > F_D$$

P4) Electric Field Mag

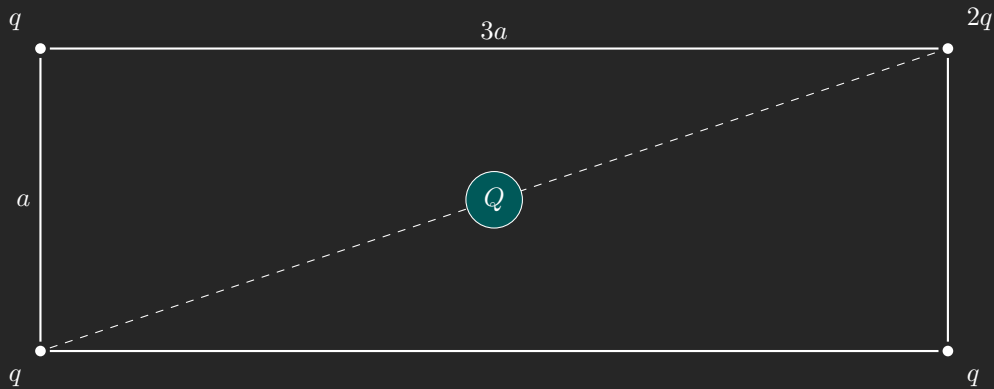
Because a test charge placed in a field exerts doesn't have any force of its own, we use a modified version of 1.3

$$E = k \frac{Q}{r_a^2}$$

Based on above...

$$2Q \propto 2E, \quad 2q \propto 2E$$

Example 2.5.2 Electric Field Due to Point Charges Four point charges $2q$, q , q , and q are placed at the corners of a rectangle of dimensions a and $3a$ as shown in the figure. A fifth charge Q is placed at the center of the rectangle. Our task is to compute the electric field at the center of the rectangle, and then determine the force on Q .



P1) Find the mag of field charges

$$r = \sqrt{\left(\frac{3a}{2}\right)^2 + \frac{1}{(2a)^2}}$$

$$r = \sqrt{\frac{9a^2}{4} + \frac{1}{4a^2}}$$

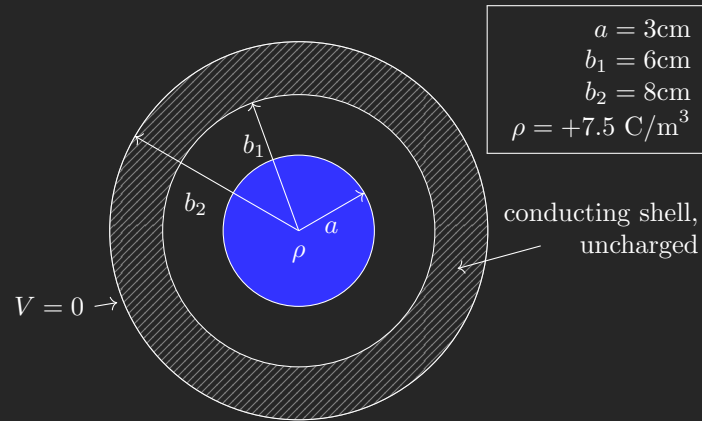
$$r = \sqrt{\left(\frac{10a^2}{4}\right)}$$

$$r = \frac{\sqrt{5}}{\sqrt{2}}a$$

$$q = E_1 = \frac{2ka}{5a^2}, \quad 2q = E_2 = \frac{4ka}{5a^2}$$

P3) Find X and Y components

Example 2.6.1 Electric Potential in a System with Cylindrical Symmetry



$$\begin{aligned} a &= 3\text{cm} \\ b_1 &= 6\text{cm} \\ b_2 &= 8\text{cm} \\ \rho &= +7.5 \text{ C/m}^3 \end{aligned}$$

Find the potential at $r = 10\text{cm}$

Find the energy enclosed using Gauss's Law (1.6)

$$\int \vec{E} dS = \frac{Q_{enc}}{\epsilon_0}$$

$$\lambda = \rho * \pi * a^2$$

$$Q_{enc} = \rho \pi a^2 L$$

$$E(2\pi r) = \rho \pi a^2$$

$$E = \frac{\rho a^2}{2\epsilon_0 r}$$

$$V(r) = \int_r^{b_2} \frac{\rho a^2}{2\epsilon_0 r}$$

$$V(r) = \frac{\rho a^2}{2\epsilon_0} \ln\left(\frac{b_2}{r}\right)$$

2.7 Equations

$$\vec{F}_{12} = k \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12} \quad (1.1) \text{ (Coulomb's Law)}$$

$$F_{net} = \sum_i \vec{F}_i \quad (1.2) \text{ (Superposition Principle)}$$

$$E = k \frac{q}{r^2} \quad (1.3) \text{ (Electric Field)}$$

$$D = \frac{N}{4\pi r^2} \quad (1.4) \text{ (Density of Field Lines)}$$

$$N \equiv \frac{q}{\epsilon_0}, \quad E = \frac{N}{A} \quad (1.5) \text{ (Number of Graphed Field Lines)}$$

$$\int E dS = \frac{Q_{enclosed}}{\epsilon_0} \quad (1.6) \text{ (Guass's Law)}$$

$$\Delta U = q \vec{E} \ell \quad (1.7) \text{ (Change in potential in const E field)}$$

$$\Delta U = Q \int_a^b \vec{E} dr \quad (1.8) \text{ (Change in potential in non static field)}$$

$$C \equiv \frac{Q}{\Delta V} \quad (1.9) \text{ (Capacitance)}$$

$$C = \frac{\epsilon_0 A}{d} \quad (1.10) \text{ (Capacitance for Parrel Plates)}$$

$$U = \frac{1}{2} \frac{Q^2}{C} \quad U = \frac{1}{2} C V^2 \quad U = \frac{1}{2} Q V \quad (1.11) \text{ (Potential of a Capacitor)}$$

$$u = \frac{1}{2} \epsilon_0 E^2 \quad (1.12) \text{ (Energy density of a capacitor)}$$

2.8 Constants

$$e = -1.6 \times 10^{-19} C \quad (1) \text{ (Electron Charge)}$$

$$k = 9 \times 10^9 \frac{NM^2}{C^2} \quad (2) \text{ (k)}$$

Glossary

Dielectric Constant Denoted as κ

$$C_{new} = \kappa C_{old}$$

The amount which the capacitance of a capacitor increases when a dielectric is placed between the charges surfaces [13](#)